

Product Specification & Operations Manual

Version 1.0 MVP — Intelligent Digital Twin for Workplace Wellbeing

Production URL: https://ai-wellness-twin.vercel.app	Date: August 2026
Repository: github.com/ronnielmharlssb-dot/ai-wellness-twin	Evaluation Stage: Stage 4 MVP

1. Executive Summary

Modern knowledge workers face chronic, hidden burnout caused by fragmented focus, excessive meeting loads, and blurred after-hours boundaries. Traditional corporate wellness initiatives fail because they rely on reactive, quarterly surveys that employees do not trust.

AI Wellness Twin is a proactive, privacy-first digital twin platform. It passively monitors work pattern telemetry across 9 essential productivity tools (IDE, GitHub, Calendar, Slack, etc.) and combines it with active MBI-GS (Maslach Burnout Inventory) reflections. Each worker is calibrated against their own 28-day baseline to predict burnout risks and simulate corrective habits before exhaustion occurs.

Bi-Directional Telemetry Passive background signals + validated subjective surveys.	k-Anonymity ($k \geq 3$) HR views group stats only for 3+ members. Zero prompt logging.	What-If Simulator Interactive parameter model simulating recovery trajectories.
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2. System Architecture & Role-Based Access Control (RBAC)

The system enforces **strict bi-directional role isolation** between employees and HR administrators:

Dimension	Employee Role (employee)	HR Admin Role (hr)
Portal Route	<code>/dashboard</code>	<code>/hr</code>
Access Boundary	Private personal twin; strictly blocked from <code>/hr</code> .	Workforce aggregate radar; strictly blocked from <code>/dashboard</code> .
Data Visibility	Personal metrics, MBI-GS scores, twin simulator, recommendations.	Anonymized group workload heatmaps, macro trends ($k \geq 3$).
Management Tools	Personal integrations, profile, notification curfews.	Department teams, single-use invite token generator.

3. Employee Experience Portal (`/dashboard`)

- **Overview (`/dashboard`):** Real-time Twin Health Score (0-100), 4-axis Burnout Risk Radar, and live Workday Signals 3x3 grid.

- **Work Pattern Diagnostics (/dashboard/patterns):** Analysis of meeting fragmentation, uninterrupted focus blocks, and after-hours activity.
- **MBI-GS Assessment Surveys (/dashboard/assessment):** Scientifically validated Maslach Burnout Inventory questions evaluating Exhaustion, Cynicism, and Efficacy.
- **What-If Twin Simulator (/dashboard/simulator):** Dynamic simulation engine forecasting burnout risk recovery based on meeting reductions and focus block adjustments.
- **Personalized Recommendations (/dashboard/recommendations):** Actionable micro-habits and smart boundary recommendations tailored to personal telemetry.
- **Integrations Manager (/dashboard/integrations):** Central management hub for OAuth connectors and background telemetry listeners across 9 apps.

4. HR Workforce Administration Portal (/hr)

- **Workforce Overview (/hr):** Organizational vitality cockpit showing macro burnout radar, average rest buffers, and workload changes.
- **Teams & Groups (/hr/teams):** Departmental management enforcing the strict $k \geq 3$ privacy threshold. Sub-groups with < 3 members are locked to protect individuals.
- **Single-Use Invite Generator (/hr/teams):** Cryptographic token generator to provision one-time onboarding invites for new company employees.
- **Longitudinal Workforce Trends (/hr/trends):** Multi-cycle trend heatmaps tracking company-wide rest patterns and overtime sustainability.

5. Supported Telemetry Integrations (9 Productivity Apps)

All integrations pass through the **Metadata Firewall**: timestamps and durations are captured, while sensitive prompt text, code snippets, and meeting titles are strictly stripped.

App / Tool	Category	Telemetry Captured (Zero Text / Prompt Logging)
GitHub	Code & Commits	Commit & PR review timestamps to measure dev windows and late-night coding.
VS Code	Engineering Focus	Active editor focus time, typing pace, and uninterrupted deep-work blocks.
ChatGPT (OpenAI)	AI Reasoning	AI consultation session duration and frequency (prompt content firewalled ■).
Google Gemini	AI Research	Research & analysis intervals and focus intensity (prompts discarded ■).
Claude (Anthropic)	AI Writing	Writing & reasoning session durations (zero prompt text logging ■).
Google Calendar	Meetings & Syncs	Meeting density, call duration, and rest buffer gaps (event titles stripped ■).
Figma	Design & Creative	Design canvas active windows and creative flow session durations.

Slack	Communication	Active messaging timestamps outside core hours to protect right-to-disconnect.
Discord	Voice & Community	Voice channel and community active windows to monitor late-night habits.

6. Privacy Architecture & Ethical Framework

- **1. k-Anonymity ($k \geq 3$):** HR administrators can never view individual employee scores. Aggregate department scores require a minimum of 3 members.
- **2. Self-Referential Baselines:** Every worker is calibrated against their own 28-day historical pattern. There are zero company leaderboards or peer rankings.
- **3. Zero Keystroke / Prompt Logging:** Telemetry is strictly metadata-only. Prompts, chat text, and source code are never parsed or stored.
- **4. Complete Role Isolation:** Employees have zero access to HR tools; HR admins are strictly prevented from viewing individual employee dashboards.

7. Test Accounts & Verification Guide

Reviewers can test both roles on the live production deployment at <https://ai-wellness-twin.vercel.app>:

Role	Email	Password	Portal URL	Key Capabilities to Test
■ Employee	ronnie@company.com	password123	/dashboard	Health score, 9-app grid, MBI-GS survey, twin simulator, weekly reports.
■■ HR Admin	hr@company.com	password123	/hr	Workforce overview radar, team $k \geq 3$ guard, single-use invite generator, trends.

■ *Tip: The /login page features instant 1-click buttons for both Employee and HR Admin for immediate evaluation.*